$$f(x)=10\sqrt[5]{x^3}-\sqrt{x^7}+6\sqrt[3]{x^8}-3$$

$$\frac{df}{dx}=\frac{df}{dx}10x^{\frac{3}{5}}-x^{\frac{1}{7}}+6x^{\frac{8}{3}}-3$$

$$\frac{df}{dx}=\frac{30}{5}x^{\frac{3}{5}-\frac{5}{5}}-\frac{1}{7}x^{\frac{1}{7}-\frac{7}{7}}+\frac{48}{3}x^{\frac{8}{3}-\frac{3}{3}}$$

$$\frac{df}{dx}=6x^{\frac{-2}{5}}-\frac{1}{7}x^{\frac{-6}{7}}+16x^{\frac{5}{3}}$$

$$\frac{df}{dx}=\frac{6}{\sqrt[5]{x^2}}-\frac{1}{7\sqrt[7]{x^6}}+16\sqrt[3]{x^5}$$